STILLSTACK · PROJECT PLAN

StillStack project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Decisions of 2026-10-25 recorded (D-04 to D-06, D-08 to D-10); coupon test scope (D-04); prototype parts USD 460.70
0.1	2026-10-24	Amish Chadha	First project plan (kit 1.9.1 upgrade), with the reference list of every source cited in the repo

StillStack project plan

StillStack is at stage P3 (design outputs, TRL 3): model detailed to D3 for both concept options, drawings, priced BOM, build plan, decisions register, design FMEA and traceability exist. The next gate is P4 (verification, TRL 4), which stays on hold under the portfolio cap. Decision requested from Amish: the concept selection D-03 (options A and B are within 10 %, sent without a pick). D-04, D-05, D-06, D-08, D-09 and D-10 were decided by Amish on 2026-10-25 (SSK-DEC-001). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and science	Design lead	STEM check SSK-STM-001, concept options SSK-OPT-001, DDR-004	Done at P3; STEM open (ten plausible rows, two fixes with Amish); selection awaiting Amish
Requirements and traceability	Design lead	SSK-URS-001, SSK-PRS-001, SSK-RTM-001	Done at P3
Calculations	Design lead	SSK-CAL-001 v0.8 (`sizing.py`, `stem_check.py`), corrected after the independent review	Done at P3
Detailed design	Design lead	`cad/src/model.py` at D3 for options A and B, STEP per part and per assembly, drawings SSK-DWG-002 and 101 to 120	Done at P3
Build plan	Design lead	SSK-BLD-001	Done at P3 (a plan, not a record)
Regulatory, risk and market	Design lead	SSK-REG-001, SSK-RSK-001, SSK-MKT-001	First pass at P3
Partner	Amish	Approach the first candidate lab (Politecnico di Torino group)	Not started; Amish's decision
Verification (P4)	Partner lab	Wicking-rise test, coupon corrosion test of 1050 or 3003 plate, wick and 5052 clip in hot brine (D-04), stake pull-out check at the site (D-09, TRL 5), stagnation test (ISO 9806 method), per-stage conductivity, outdoor yield beside a basin still	On hold (TRL 3 cap)

Resources needed

Table 2. Resources (P4 and later on hold; costs are not estimated until the cap is lifted).

Resource	Deliverable	Effort	Estimated cost (USD)
Prototype parts	One panel and stand (BOM)	One builder, about a week	460.70 (BOM)
Partner lab time	Distillate quality and yield tests	To assess with the lab	To assess
Coater	Certified food-contact coating on four plates	One batch	About 20 in the BOM; certificate cost to assess
Field partner	Household trial (P5)	To assess	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Done (2026-10-24)
P1 Definition	URS, PRS, RSK with hazard analysis, MKT	Done (2026-10-24)
P2 Design inputs	PRS targets and verification, CAL, RTM, concept selection DDR	Done; the selection itself awaits Amish (D-03)
P3 Design outputs	Model at D3, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Done (2026-10-24)
P4 Verification	Test plans and reports, verification report	On hold (TRL 3 cap)
P5 Validation	Field trial with a partner, certification status	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Amish decides D-03 (concept selection); D-04 to D-06 and D-08 to D-10 were decided on 2026-10-25 (SSK-DEC-001).
2. The coordinator renders `media/options.png` and the product views in `RENDER_VIEWS` in Blender Cycles on Amish's Mac.
3. When the cap is lifted: approach the partner lab, then the TRL 4 tests listed above, starting with the wicking-rise test (it decides PR-001 and PR-006).

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